

NEXORA

Intelligent Business Operations Portal

Full Stack Mini ERP + CRM Case Study Submission for FundsRoom

1. Executive Summary & Live Project Links

GitHub Repository:	https://github.com/ChaithanyaSakamuri/FundsRoom-Assignment
Live Frontend URL:	https://nexora-frontend-papj.onrender.com/
Live Backend API URL:	https://nexora-backend-zzs8.onrender.com/
Author / Candidate:	Chaithanya Sakamuri
Deployment Platform:	Render Cloud PaaS (Web Services & Global Static CDN)

2. Test Login Credentials for All Roles

NEXORA features a granular Role-Based Access Control (RBAC) system. Use the following credentials on the login screen or click the corresponding pre-filled role buttons:

Role	Email Address	Password	Access Scope & Permissions
Admin	admin@nexora.demo	nexora@2024	Full Unrestricted Access
Sales	sales@nexora.demo	nexora@2024	CRM, Challans, Sales Reports
Warehouse	warehouse@nexora.demo	nexora@2024	Inventory, Stock Ledger
Accounts	accounts@nexora.demo	nexora@2024	Challan Invoices, Reports

3. System Architecture & Core Design Principles

NEXORA is engineered as a modern, decoupled full-stack enterprise B2B SaaS application. It combines real-time operational state management with transactional stock ledger integrity.

Tech Stack Overview:

- **Backend API:** Node.js 20, Express.js framework, TypeScript, Prisma ORM, JWT authentication
- **Frontend SPA:** React 18, Vite, TypeScript, TanStack Query (React Query), Zustand, Recharts, Lucide Icons
- **Design System:** Vanilla CSS with curated design tokens, glassmorphism, 3-layer dark mode, responsive drawer
- **Document Engine:** PDFKit for

dynamic server-side official
delivery challan PDF generation

- **Data Storage:** Prisma ORM with cross-database support (PostgreSQL & SQLite file database)

Key Architectural Highlights:

- **Atomic Stock Fulfillment:** When confirming a delivery challan, Prisma executes a multi-table transaction that checks stock levels, decrements inventory, creates stock movement logs, and confirms the order atomically.

- **Historical Snapshot Preservation:** Challan line items store immutable JSON snapshots of product price, unit, and specs at order time, protecting audit history from future price modifications.

- **Global Command Palette:** Ctrl+K / # K command overlay with live multi-entity search across customers, catalog items, and sales orders.

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4. Complete API Endpoint Documentation

The NEXORA REST API exposes clean, versioned endpoints requiring Bearer JWT authentication for protected routes:

Method	Endpoint Route Path	Description & Business Function
POST	/api/auth/login	Authenticate user & issue JWT token
GET	/api/auth/me	Fetch authenticated user profile
GET	/api/customers	List CRM customers with search, status & pagination
POST	/api/customers	Create new customer account [Sales/Admin]
GET	/api/customers/:id	Customer 360° profile with timeline & challans
GET	/api/products	List inventory catalog with stock health indicators
POST	/api/products	Add new product SKU [Warehouse/Admin]
GET	/api/stock-movements	Stock audit ledger (filter by IN/OUT type & date)
POST	/api/stock-movements	Record manual stock adjustment [Warehouse/Admin]
GET	/api/challans	List sales delivery challans with filter & pagination
POST	/api/challans	Create 5-step draft sales delivery challan
POST	/api/challans/:id/confirm	Confirm order & execute atomic stock deduction
GET	/api/challans/:id/pdf	Generate & stream official PDF invoice document
GET	/api/dashboard/summary	Command center KPIs, trend charts & alerts
GET	/api/reports/sales-overview	Analytics report for sales revenue velocity
GET	/api/activity	System-wide audit trail activity log

5. Setup & Deployment Guide

Local Environment Setup:

- **1. Clone Repository:** git clone <https://github.com/ChaithanyaSakamuri/FundsRoom-Assignment.git>
- **2. Backend Setup:** cd backend && npm install && npx prisma db push && npm run db:seed && npm run dev
- **3. Frontend Setup:** cd frontend && npm install && npm run dev (opens <http://localhost:5173>)

Render Cloud Deployment:

- **Render Blueprint:** Included render.yaml configures 1-click deployment for both backend & frontend static site.
- **Backend Web Service:** Build: npm install && npx prisma generate && npm run build | Start: npx prisma db push && npm start
- **Frontend Static Site:** Build: npm install && npm run build | Environment Variable: VITE_API_URL pointing to live backend.

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6. Known Scope Decisions & Trade-offs

- **Single JWT Session:** Uses 24-hour JWT access tokens. Production refresh token rotation pattern can be attached for multi-device session revocation.
- **Image Storage:** Product catalog stores image URLs. Production S3/Cloudinary direct upload widget can be added for binary asset hosting.
- **Real-Time WebSockets:** UI updates on React Query polling/revalidation. WebSocket layer can be added for multi-user live stock updates.

7. Project Sign-Off

NEXORA has been thoroughly built, tested, and deployed according to all requirements outlined in the Full Stack Developer Case Study. The application demonstrates production-level engineering quality, clean code architecture, and extreme visual polish.

Project Submission Verification — COMPLETE

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